SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Short Answer

Course Code:	ITA03	**Course Title:**	Mobile computing
CO Assessed:	CO1	**Assessment:**	Assessment Tool 2 – Short Answer
Weightage:	30 % of CIA	**Total Marks:**	100 Marks
CO1 Statement:	*Analyze the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.*		
Student Name:	J. Srilekha	**Reg. No.:**	192821129.
Section / Batch:	30 - 7 - 2026	**Date:**	Slot D.

Code of Conduct

I J. Srilekha (192321129) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: J. Srilekha

Instructions

- This test contains 50 questions, Total: 100 Marks.
- Duration: 1 hour.

85/100

Questions

Q1. A telecom operator wins a spectrum auction for 45 MHz. Company policy sets aside 25 kHz for every simplex channel. The planning team wants to know how many full-duplex voice channels this spectrum can support.

Q2. A regional carrier runs its network on 560 total voice channels, shared across a 7-cell frequency reuse cluster. During capacity planning, an engineer needs to know how many channels a single cell will typically be allocated.

Q3. A GSM operator sets carrier spacing at 200 kHz and rolls out 100 carriers, each carrying 8 time slots for voice traffic. Determine the total number of simultaneous voice channels this deployment can offer.

Q4. During a site audit, engineers note the base station radiates 80 W while a nearby handset transmits just 4 W. They want to express this power gap as a decibel-based power control ratio (10·log10 of the ratio).

Q5. A drive test captures a serving signal at -60 dBm alongside co-channel interference at -92 dBm. The RF team needs the Carrier-to-Interference ratio (C/I) in dB to assess link quality.

Q6. An operator holds 42 MHz of spectrum, assigning 300 kHz per channel, and reuses frequencies across a 7-cell pattern. How many channels does a single cell end up with?

Q7. A transmitter delivers 64 W of power to a receiver. Engineers then move the receiver twice as far away, in an environment with a path-loss exponent of 4. What received power should they now expect?

Q8. Planners want to double a cell's radius while keeping the same SNR at the new edge, in an environment obeying an inverse-square power law. By what factor should transmit power be scaled up?

Q9. A network is built with a cluster size of 4, and every cell in the cluster is provisioned with 15 voice channels. What is the total channel count across the entire cluster?

Q10. A drive test records a vehicle traveling at 100 km/h while camped on a 900 MHz carrier. Estimate the Doppler shift the receiver experiences.

Q11. A TDMA frame lasting 5 ms is divided into 8 equal time slots. What is the duration of a single slot?

Q12. Spectrum planners have 24 MHz of allocated bandwidth and fix carrier spacing at 200 kHz for a GSM rollout. How many carrier frequencies can be created from this allocation?

Q13. A handset transmits at 8 W, but the signal experiences 33 dB of path loss on its way to the base station. Estimate the power the base station receiver actually sees, in dBm.

Q14. A network engineer uses 1 m as the reference distance while assessing a 1900 MHz deployment, where cell sites are spaced 2.5 km apart. Estimate the free-space path loss over that distance.

Q15. A cluster consists of 6 cells, each able to carry up to 250 simultaneous calls at peak. What is the cluster's total call-handling capacity?

Q16. A switching subsystem is rated to process 4000 new call requests every second, and calls last 3 minutes on average. How many calls could be active in the system at any one time?

Q17. Engineers assign 150 channels to a base station subsystem, each channel occupying 25 kHz. What total bandwidth does this base station consume?

Q18. Spectrum planners insert a 40 kHz guard band between each of 30 adjacent frequency bands to prevent interference. How much total bandwidth ends up wasted because of these guard bands?

Q19. A vehicle cruises at 90 km/h through a region of 1.5 km-radius cells. Using the cell diameter as the crossing distance, roughly how many handoffs would occur over an hour of travel?

Q20. A carrier's network hierarchy has 8 regional centers, each overseeing 40 base stations, with each base station serving 250 users. What is the network's total user-handling capacity?

Q21. Coverage planners want to blanket a 400 sq km region using hexagonal cells of 400 m radius (approximate cell area as R^2). Roughly how many cells will the rollout require?

Q22. An FDMA system allots 30 kHz to each channel, drawing from a total spectrum pool of 6 MHz. How many channels can be carved out of this allocation?

Q23. A market study notes that 1200 subscribers each place 4 calls per hour, averaging 2.5 minutes per call. What total offered traffic, in Erlangs, does this represent?

Q24. A 2.5 GHz deployment has 40 MHz of bandwidth, and each connected user consumes 400 kHz. What is the maximum number of users the band can serve at once?

Q25. A carrier's base stations each cap out at 600 simultaneous users, and the network comprises 1200 such stations. What total user capacity does the network offer?

Q26. A paging scheme sweeps through 4 cells at a time across a 200-cell network. In the worst case, how many paging rounds are needed to locate a subscriber?

Q27. A core network is engineered for 100,000 simultaneous connections. During a busy period it receives 40,000 call attempts per minute, each lasting 3 minutes on average. What fraction of the offered traffic ends up blocked?

Q28. A network policy lets 15% of its 600,000 subscribers roam onto partner networks each hour. At any given moment, how many subscribers are likely roaming?

Q29. A switching center is capable of processing 12,000 calls per second, but is currently handling a peak load of 9,000 calls per second. What percentage of its capacity is being utilized?

Q30. An operator reuses frequencies with a factor of 4 across a network holding 48 MHz of spectrum, with 200 kHz per channel. How many channels land in each cell?

Q31. A 900 MHz GSM deployment uses 200 kHz channel spacing across a 30 MHz allocation. How many channels can this spectrum support?

Q32. Over the course of an hour, a network logs 2500 handoff attempts, 99% of which succeed. How many calls were successfully handed off?

Q33. A mobile handset radiates 12 W, and the link experiences 45 dB of path loss before reaching the base station. What power level does the base station receiver detect?

Q34. A 7-cell cluster is deployed where each cell can serve up to 900 simultaneous users. What is the cluster's overall user capacity?

Q35. A TDMA carrier's frame spans 4.8 ms and is split into 8 equal slots. How long is each individual slot?

Q36. A mobile device moves at 150 km/h while connected to a 1.8 GHz carrier. What Doppler shift should the receiver expect?

Q37. A core switch is dimensioned for 250,000 simultaneous calls. It is fielding calls at a rate of 70,000 per minute, each averaging 4 minutes in duration. What share of the offered load gets blocked?

Q38. A hexagonal layout groups 9 cells into a cluster, and each cell can host 600 users. What is the cluster's combined user capacity?

Q39. A switching center processes 60,000 calls per second, with 0.4% of them dropped due to network issues. How many calls are being dropped every second?

Q40. A base station in an 1800 MHz network transmits at 43 dBm. At the edge of its 2 km-radius cell, the received signal measures -87 dBm. What path loss has the signal undergone?

Q41. A single base station covers 2.5 km^2 of territory, and the operator plans to deploy 150 identical stations. What total geographic area will the rollout cover?

Q42. An operator applies a frequency reuse factor of 6 over a 48 MHz spectrum allocation, with each channel taking up 200 kHz. How many channels does each cell get?

Q43. A paging system processes 6000 users per cycle, and the subscriber base totals 150,000 users. How many paging cycles are needed to sweep the entire base?

Q44. A phone call runs for 9 minutes, and during that time the network hands the call off roughly every 3 minutes. How many handoffs took place over the call?

Q45. A receiver picks up a wanted signal at -85 dBm against a noise floor of -115 dBm. What signal-to-noise ratio, in dB, does this represent?

Q46. A 4G cell is serving 250 users simultaneously, each needing a guaranteed 150 kbps. What is the minimum bandwidth the cell must provide to support them all?

Q47. A base station handles 1500 subscribers, each averaging 4 calls per hour with a 2.5-minute call length. What total traffic, in Erlangs, does this generate?

Q48. A network reserves 250 separate control channels, each 12 kHz wide, for signaling. What total bandwidth is tied up in control channels?

Q49. A network is carrying 1500 simultaneous connections, each streaming at 600 kbps on average. What total throughput must the network support?

Q50. A telecom operator wants each tower to cover a 4 km radius (treating coverage as a circle, area = πR^2) and needs to blanket a 350 km² service region. Roughly how many towers will the rollout require?

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

Faculty In-charge	**Course Coordinator**	**Head of Department**
Signature: ____________	Signature: ____________	Signature: ____________
Date: ____________	Date: ____________	Date: ____________

Mobile Computing

900 channels (45 MHZ/50 KHz per full-duplex channels since each full - duplex channel require 100 25 KHZ simplex channel.

80 channels (560 total channels /7 cells)

800 channels (200 carrier x 8 time slots)

13 dB ($20 \cdot \log_{10}(80/4) = 10 \cdot \log_{10}(20 = 13)$)

32 dB (-60 dBM - (-92 dBM) = 32 dB)

20 channels per cell (Total channels = 42 MHz / 300 kHz = 140 channels ; per cell = 140/7).

40 ($P_r \propto d^{-4}$; doubling d reduces P_r by 4, so P_t must 10/16 = 4.0)

(8) 16 time ($P_r \propto P_t \cdot d^{-2}$, doubling d reduces P_r by 4 so P_t must increase by $2^4 = 16$ for the same SNR.

(9) 60 channels (4 cells × 15 channels)

(10) 83.3 Hz ($f_2 = \frac{V}{\lambda} \cdot \cos\theta$; V = 27.77 M/s , λ = 0.333 M

(11) 0.625 MS (5 MS/8)

(12) 120 carriers (24 MHz/200 KHz)

(13) 6.03 dBM (8 W = 39 dBM ; 39 dBM . 3 dB = 6 dBM)

(14) 206.012 B

(15) 1,500 calls (6 cells × 250 calls).

(16) 720,000 active calls (4000 calls / sec × 180 sec.)

(17) 3.75 MHz (150 × 25 kHz)

(18) 1.26 MHz (29 guard bands × 40 kHz)

(19) 30 handoffs (d = 3 km; distance in 1 hr = 90 km; 90/3 = 30)

(20) 80,000 users (8 × 40 × 250)

(21) 2,500 cells (total area 400 km^2 / cell area 0.16 km^2 where Area

$R^2 = 0.4^2 = 0.16\ km^2$.)

(22) 200 channels (6 MHz / 30 kHz)

(23) 2000 Erlangs

(24) 100 user.

(25) 720,000 users {(600 users / station × 1200 stations)

(26) 50 paging round (200 cells / 4 cells per round)

(27) 16.67% or 1/6 (offered traffic = 120,000 Erlangs)

(27) 90,000 subscriber (600,0000 × 15%).

(29) 75% (9000 / 12,000)

(30) 60 channels (total channels = 48 MHz / 200 MHz = 240)

(31) 150 channels

(32) 2,475 calls (2500 × 0.99)

(33) -4.21 dBM

(34) 6,300 users

(35) 0.6 ms (4.8 ms / 8)

... Hz ($f_d = \frac{v \cdot f}{c}$ where $v = 41.67 M/s$, and $f = 1.8 GHz$)

10.71% or 3/28 (offered load = 280,000 Erlangs, capacity = 250,000; share blocked = 280,000 − 250,000/280,000)

38) 5,400 users (9 cells × 600 users/cell)

39) 240 calls (60,000 × 0.004)

40) 130 dB (42 dBm − (−87 dBm))

41) 375 km² (2.5 km² × 150 stations)

42) Each cell gets 40 channels

43) 25 paging cycles are needed

44) 2 handoffs took place

45) The SNR is 30 dB

46) Minimum bandwidth is 37.5 Mbps

47) Total traffic generated is 250 Erlangs

48) Total bandwidth tied up is 3 MHz

49) Total supported throughput must be 900 Mbps

50) The rollout will require flowers